Demand Estimation with Unstructured Product Data

Evidence from Amazon's Toothpaste Market

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Abstract

Merger simulation hinges on correctly measuring substitution patterns, yet the nesting structures used in practice are typically built from coarse labels such as brand identity. This paper asks whether product representations extracted from unstructured listing content – images and text encoded with CLIP (Radford et al. 2021) – define substitution patterns better than brand labels do. Using an ASIN-by-quarter panel of the Amazon US toothpaste market around Colgate-Palmolive's 2020 acquisition of the natural-brand hello, I estimate three nested-logit demand models via the Berry (1994) linearisation that differ only in how nests are assigned: no nesting, brand nesting, and nesting by k-means clusters of CLIP image-and-text embeddings. CLIP-based nesting fits demand best and, critically, reduces the share of economically implausible negative implied marginal costs from 8.6% to 1.7%. Because CLIP places hello and Colgate in different clusters, the implied diversion between the merging parties is small: the simulated merger price effect for hello falls from +1.23% under plain logit to +0.40% under CLIP nesting. The results suggest embedding-based nests are a practical, replicable tool for antitrust analysis when conventional product characteristics are unavailable.

1 Introduction

Antitrust authorities increasingly confront mergers in digital markets – characteristically, an established incumbent acquiring a fast-growing, digitally native brand – in which product differentiation is not described by standardised attributes. In a car merger the analyst observes horsepower and fuel economy; in a laptop merger, RAM and screen size. In an online consumer-goods merger, the economically relevant differentiation – a "natural" positioning, sensitivity relief, visual branding, a kid-friendly format – lives in unstructured listing content: product images, titles, and free-form benefit descriptions. The workhorse tools of merger simulation, however, were built for the structured world. Nested-logit models encode substitution through nest assignments constructed from coarse observable labels, most commonly brand identity or hand-picked segments, and when those labels misrepresent how consumers perceive products, the resulting diversion ratios, markups, and simulated price effects inherit the error.

A second, equally practical obstacle compounds the first. Estimating price sensitivity credibly requires exogenous cost-side variation, and high-frequency online retail data rarely contain any: input-cost series vary only over time and are absorbed by time fixed effects, while Hausman-style rival-price instruments fail when demand shocks are correlated across markets, as they plainly are on a single national platform. The symptom of proceeding without valid instruments is well known to practitioners: demand estimated far too inelastic, implying Bertrand markups so large that the marginal costs recovered from the first-order conditions turn *negative* for a substantial share of products. This negative-cost rate is not a cosmetic nuisance – it is a direct, observable measure of how badly a demand system describes competition, and it is the diagnostic this paper puts at the centre of the analysis.

This paper offers a pragmatic pipeline for exactly that environment, with one methodological contribution at its core. The pipeline maps unstructured product listings into a continuous differentiation space using a pre-trained deep learning model – CLIP image and text embeddings (Radford et al. 2021), reduced to five joint principal components – and sidesteps the missing-instrument problem by calibrating the price coefficient to an established industry elasticity, the meta-analytic mean own-price elasticity of -2.62 for consumer packaged goods (Bijmolt, van Heerde, and Pieters 2005). The contribution lies in what the embeddings are used *for*: nests are assigned by k-means clustering in the embedding space rather than by brand. Holding everything else fixed, I compare three Berry (1994) nested-logit models that differ only in their nest assignment – no nesting, brand nesting, and CLIP-cluster nesting – and let the negative-cost diagnostic adjudicate between them.

The empirical laboratory is Colgate-Palmolive's acquisition of hello Products LLC, a fast-growing premium "naturally friendly" oral-care brand. The deal was announced on 23 January 2020 and completed on 31 January 2020 for $351 million in cash, giving a clean event date (2020Q1) inside the sample. Using an ASIN-by-quarter panel of the Amazon US toothpaste market (2018Q1–2022Q3, 159 products), I estimate three demand models via the Berry (1994) nested-logit linearisation that are identical in every respect except the nest assignment: M1 imposes no nesting (plain logit), M2 nests by brand, and M3 nests by K=5 CLIP clusters. With the price coefficient fixed by calibration, the remaining parameters are estimated by concentrated OLS with quarter fixed effects and the nesting parameter ρ is profiled by grid search. Marginal costs are recovered from the Bertrand first-order conditions, and the merger is simulated by recomputing Nash equilibrium prices under pre- and post-merger ownership at fixed pre-merger costs.

Three results stand out. First, CLIP-based nesting fits demand best: the profiled nesting parameter is $\rho^* = 0.67$ and the residual sum of squares falls by 8.1% relative to plain logit, against 3.1% for brand nesting. Second – and this is the central finding – CLIP nesting drives the share of observations with *negative implied marginal costs* from 8.6% under both plain logit and brand nesting to 1.7%. Negative recovered costs are the canonical symptom of a demand system that understates substitutability and therefore overstates markups; their near-elimination is direct evidence that embedding-based nests capture the competitive structure that brand labels miss. Third, the economics of the merger change materially: CLIP places hello and Colgate in different clusters, implying low diversion between the merging parties, and the simulated price effect for hello falls from +1.23% (plain logit) to +0.40% (CLIP nesting), with Colgate's own effect essentially zero.

Beyond the specific merger, the exercise is designed as a transparent, fully reproducible template for antitrust practitioners: every step – from embedding construction through the ρ grid search to the Nash counterfactuals – runs from a single document, and the Nash solver is validated against PyBLP's counterfactual routines (Conlon and Gortmaker 2020) to numerical precision. The remainder of the paper proceeds as follows. Section 2 relates the approach to the literature. Section 3 describes the industry setting, the data, and the embedding construction. Section 4 lays out the empirical framework. Section 5 presents the results, and Section 6 discusses limitations. Section 7 concludes.

2 Related Literature

The demand framework descends from the discrete-choice literature on differentiated products. Berry (1994) shows that logit and nested-logit demand can be estimated by linear methods after inverting market shares, and that the same first-order conditions used to rationalise observed prices can be inverted to recover marginal costs under Bertrand competition. Nevo (2000) provides the canonical practitioner's treatment of this class of models, and Nevo (2001) demonstrates the full structural program – demand estimates, recovered margins, and competitive counterfactuals – in the ready-to-eat cereal industry, a consumer packaged goods setting closely analogous to toothpaste. Conlon and Gortmaker (2020) codify current best practice and provide the PyBLP routines against which this paper's equilibrium computations are validated. Draganska and Jain (2006) estimate structural demand for differentiated CPG product lines and report own-price elasticities of the magnitude used for calibration here.

Within this tradition, the paper's price coefficient is calibrated rather than instrumented. The target elasticity of -2.62 comes from Bijmolt, van Heerde, and Pieters (2005), who meta-analyse 1,851 own-price elasticity estimates from 81 studies and report -2.62 as the mean. Calibration to an external elasticity is standard in applied antitrust simulation when credible cost-side instruments are unavailable, and it has a transparency advantage in a practitioner context: the price-sensitivity assumption is stated in one number rather than buried in a first stage.

The use of unstructured data is most closely related to Compiani, Morozov, and Seiler (2026), who extract embeddings from product images and text with pre-trained deep learning models and incorporate them into a mixed logit model, showing across 40 Amazon categories that unstructured data identify close substitutes that attribute-based models miss. This paper shares their premise – embeddings encode the differentiation consumers care about – but differs in implementation and aim. Where Compiani, Morozov, and Seiler (2026) enter embeddings as continuous random-coefficient characteristics in a mixed logit, I use them to *define the nesting structure* of a Berry (1994) nested logit. The cost of this simplification is a coarser substitution pattern; the benefit is that the entire pipeline remains linear, transparent, and estimable in seconds, which matters for the merger-review setting where speed and auditability are first-order concerns. The embedding technology itself, CLIP, is due to Radford et al. (2021); its contrastive image-text training makes image and text representations directly comparable, which is what allows a single joint product space to be built from both modalities.

The contribution, then, sits at the intersection: a minimal, replicable bridge between modern representation learning and the nested-logit merger simulation toolkit that antitrust practitioners already use.

3 Industry Background and Data

3.1 The hello–Colgate Acquisition

hello Products LLC, founded in 2012, positioned itself as a "naturally friendly" oral-care brand – fluoride-optional formulas, charcoal and coconut-oil lines, playful branding – with strong appeal among younger consumers. Colgate-Palmolive announced its agreement to acquire hello on 23 January 2020 and completed the transaction on 31 January 2020 for $351 million in cash. The acquisition is an attractive empirical laboratory for three reasons: the event date falls cleanly at the start of 2020Q1; the acquirer is the category's largest incumbent while the target occupies a differentiated "natural" niche, so the price effect hinges on exactly the substitution question this paper studies; and both parties sell extensively on Amazon, where listing content is observable.

3.2 Data Sources

The analysis combines two data sources. Household-level purchase histories come from the Open E-commerce 1.0 dataset (Berke et al. 2024), a crowdsourced panel of 5,027 US Amazon consumers with more than 1.8 million transaction records, distributed through Harvard Dataverse. The dataset has been validated by its authors against public Amazon sales data: aggregate expenditure in the panel correlates with published Amazon revenues at Pearson $r = 0.978$ ($p < 0.001$; Berke et al. 2024). The toothpaste purchase records extracted for this study span January 2018 to March 2023 and include order timestamps, prices, and household identifiers; household demographics are available in the source data but enter the analysis only through the count of unique purchasing households per cell. Product-level metadata – titles, benefit descriptions, and image URLs for every purchased ASIN – were collected via Keepa and parsed with a rule-based extraction script (brand matching against a curated brand list, keyword-based benefit extraction, and unit-aware package-size parsing).

3.3 Sample Construction

The estimation sample is built in three steps, each dictated by a practical constraint:

1. **Choice-set restriction (167 ASINs).** The universe of toothpaste ASINs is restricted to products with at least five recorded purchases over the sample window. An unrestricted choice set with thousands of rarely purchased alternatives proved computationally infeasible and produced singular design matrices; the five-purchase floor keeps every product with meaningful revealed-preference information while discarding noise listings.

2. **Brand grouping (12 groups).** Brands accounting for more than 2% of purchases are treated as their own groups – Colgate, Crest, SENSODYNE PRONAMEL, Sensodyne, Tom's of Maine, Arm & Hammer, hello, Parodontax, APAGARD, JASON, and Orajel – and all remaining small brands are collapsed into an "Other" category. Together the eleven named groups account for 87% of purchases.

3. **Panel filters (159 ASINs, 1,164 observations).** Purchases are aggregated to ASIN x quarter cells. The estimation window ends in 2022Q3 (later quarters are incompletely covered), and ASINs must appear in at least three quarters to enter the sample, eliminating products whose near-zero shares would be dominated by sampling noise. The resulting panel covers 2018Q1–2022Q3: 19 quarters, 159 ASINs, and 1,164 observations.

Separately from these sample filters, the joint embedding space of Section 3.4 is *fitted* on the 155 of 167 ASINs that hold both a valid image and a valid text embedding; the 12 ASINs with text-only embeddings are projected into that space and remain in the estimation sample throughout. Table 1 reports summary statistics for the estimation panel.

Table 1: Panel Summary Statistics (Estimation Sample)

Variable	Mean	Median	SD	Min
Price (USD)	9.20	8.47	5.34	1.35
Quarterly units sold	2.17	1.00	1.96	1.00
Market share (x1,000)	1.093	0.753	1.036	0.335
Package size (g)	320	259	223	43
Purchasing households per cell	2.06	1.00	1.85	1.00

Notes:
1,164 ASIN x quarter observations; 159 ASINs; 19 quarters (2018Q1–2022Q3); 12 brand groups. Price is the quarterly mean deflated

3.4 Market Definition

Market shares are constructed from revealed-preference purchase counts. Market size M_t is set to $(1+\lambda)\cdot\sum_j q_{jt}$ with $\lambda = 13.93$, implying an outside-good share of approximately 93%.

3.5 From Listings to Nests: Embeddings, Joint PCA, and Clustering

Each product j is represented by its listing image and the concatenation of its title and benefit description. CLIP (Contrastive Language-Image Pre-training; Radford et al. 2021) provides an image encoder f_I and a text encoder f_T, trained contrastively so that matching image–text pairs lie close together in a shared representation space. Both encoders (`clip-vit-base-patch32`) map into $\mathbb{R}^{512}$, and outputs are L2-normalised:

$$\mathbf{e}_j^I = \frac{f_I(\text{image}_j)}{\|f_I(\text{image}_j)\|}, \qquad \mathbf{e}_j^T = \frac{f_T(\text{text}_j)}{\|f_T(\text{text}_j)\|}.$$

Each modality's embeddings are first reduced to five modality-specific principal components (Hotelling 1933), with each PCA fitted only on the products holding a valid embedding for that modality. This yields a ten-dimensional characteristics vector per product, $\mathbf{x}_j = (x_{j1}^I, \ldots, x_{j5}^I, x_{j1}^T, \ldots, x_{j5}^T)$.

Because these ten columns come from two separate PCA fits, their scales are not comparable: within each modality the first component mechanically carries the largest variance, and the image and text blocks have different overall variance levels. The joint PCA is therefore computed on the correlation scale. Each column k is z-scored using its mean μ_k and standard deviation σ_k over the $N = 155$ products with both modalities,

$$z_{jk} = \frac{x_{jk} - \mu_k}{\sigma_k},$$

and the rotation matrix $\mathbf{W}$ is obtained from the PCA of the standardised matrix $\mathbf{Z}$. The first five joint components, $\mathbf{pc}_j = \mathbf{z}_j'\mathbf{W}_{[\cdot,1:5]}$, define the differentiation space used throughout the paper.

Standardisation is not cosmetic: without it, the joint solution would be dominated by whichever input columns happen to carry the largest raw variance – the two first-stage PC1s – rather than by genuine cross-modality structure. The fitted parameters $(\mu_k, \sigma_k, \mathbf{W})$ are saved with the replication materials, which makes the projection of any product into the joint space exactly reproducible.

A useful diagnostic for whether the joint space actually blends the two modalities is the modality share of each component: the fraction of its squared rotation loadings attributable to image inputs,

$$\text{share}_m^I = \frac{\sum_{k \in I} w_{km}^2}{\sum_k w_{km}^2}, \qquad m = 1, \ldots, 5.$$

As Figure 1 shows, every retained joint component draws between 48.6% and 51.4% of its loading mass from each modality – the joint space is genuinely multimodal rather than a relabelled image space or text space.

Finally, nests are formed by k-means clustering (Lloyd 1982) on the five joint components: each product is assigned to one of $K = 5$ clusters,

$$g_j = \arg \min_{g \in \{1,\ldots,K\}} \|\mathbf{pc}_j - \mathbf{c}_g\|^2,$$

with centroids $\mathbf{c}_g$ from 50 random restarts (seed 42), and cluster membership is the nest assignment in M3. The five clusters explain 71.2% of joint-PC variance. Appendix A reports cluster-quality diagnostics – within-cluster sum of squares and average silhouette width (Rousseeuw 1987) – together with all downstream results for $K = 2, \ldots, 10$.

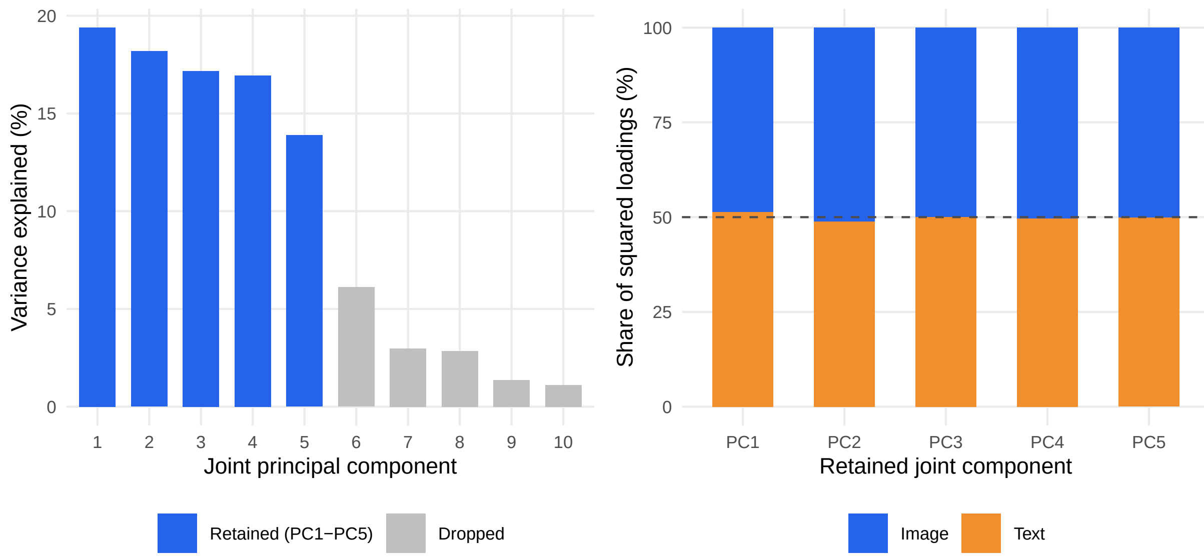

Figure 1: Anatomy of the joint embedding space. Left: variance explained by each joint principal component of the standardised ten-column input (five image PCs and five text PCs), fitted on the 155 ASINs with both modalities; the first five components (blue) are retained. Right: modality composition of each retained component, measured as the share of squared rotation loadings attributable to image versus text inputs. The dashed line marks an even 50/50 split; every component lies within 1.4 percentage points of it, confirming the joint space blends both modalities rather than being dominated by either.

Embeddings are static over the sample window. Established Amazon listings change their images and titles only marginally over 2018–2022, so re-extracting embeddings per quarter would produce near-identical vectors; the identifying variation is **cross-sectional** — ASINs within the same brand differ genuinely in packaging and benefit claims — and this is exactly the variation recovered by moving from brand-level to ASIN-level data. Time-varying demand conditions are absorbed by quarter fixed effects.

Twelve ASINs lack an image embedding (image retrieval failed at extraction time) but have valid text embeddings. Rather than dropping them or imputing brand means, they are projected into the joint PC space from their standardised text components, with image dimensions set to the cross-sectional mean – zero in standardised space, so that $\mathbf{pc}_j = (\mathbf{0},\, \mathbf{z}_j^T)' \mathbf{W}_{[\cdot,1:5]}$ using the same saved $(\mu_k, \sigma_k, \mathbf{W})$. These ASINs span Crest (4), Sensodyne (2), Tom's of Maine (2), hello (2), and Other (2), and all clear the three-quarter inclusion filter. Dropping them would distort brand-level market shares and remove hello observations inside the merger simulation window; brand-mean imputation was rejected because it would reintroduce precisely the aggregation that the ASIN-level design is meant to avoid.

4 Empirical Framework

4.1 Berry (1994) Linearisation

All three models are estimated via the Berry (1994) nested-logit linearisation:

$$\log \frac{s_{jt}}{s_{0t}} - \rho \, \log s_{j|g,t} = \alpha \, p_{jt} + \boldsymbol{\beta}' \mathbf{x}_{jt} + \mathrm{FE}_t + \xi_{jt}$$

where $s_{j|g,t} = s_{jt} / \sum_{k \in g} s_{kt}$ is the within-nest share, $\rho \in [0, 1)$ is the nesting intensity, and FE_t are quarter fixed effects (absorbed by within-quarter demeaning).

4.2 Price Coefficient Calibration

Valid cost-side instruments for price are unavailable in this high-frequency online retail dataset (see Section 5). Following the recommendation of Berry (1994) and standard practice in applied antitrust simulation, α is calibrated to match a target median own-price elasticity:

$$\alpha = \frac{\varepsilon_{\text{target}}}{\text{median}(p_{jt}(1 - s_{jt}))} = \frac{-2.62}{8.4502} = -0.310$$

The target $\varepsilon_{\text{target}} = -2.62$ corresponds to the brand-level CPG meta-analysis mean of Bijmolt, van Heerde & Pieters (2005), consistent with structural BLP estimates for analogous packaged-goods categories (Nevo 2001; Draganska & Jain 2006).

Sequential calibration note. Since α is calibrated at $\rho = 0$, the target elasticity -2.62 is the logit median. At the profiled nesting parameters ($\rho^* = 0.35$ for M2, $\rho^* = 0.67$ for M3), the nested-logit formula amplifies own-price sensitivity by approximately $1/(1 - \rho^*)$, yielding implied median elasticities of approximately -4.0 (M2) and -7.9 (M3).

4.3 ρ^* Selection — Concentrated OLS Profile

For each ρ on a fine grid $\{0, 0.01, \ldots, 0.90\}$, the adjusted dependent variable is formed and $\boldsymbol{\beta}(\rho)$ is estimated by OLS. The profiled ρ^* minimises the residual sum of squares:

$$\tilde{y}_{jt}(\rho) = \log \frac{s_{jt}}{s_{0t}} - \alpha p_{jt} - \rho \log s_{j|g,t}, \qquad \rho^* = \arg\min_{\rho} \left\| \tilde{y}(\rho) - \mathbf{X} \hat{\boldsymbol{\beta}}(\rho) \right\|^2$$

4.4 Marginal Costs and Merger Simulation

Marginal costs are recovered from the pre-merger Bertrand first-order condition (Berry 1994):

$$\mathbf{mc} = \mathbf{p} + (\Omega_{\text{pre}} \circ \Delta)^{-1} \mathbf{s}$$

where $\Omega[j, k] = \mathbf{1}[\text{firm}_j = \text{firm}_k]$ and Δ is the demand Jacobian with elements:

$$\Delta_{jj} = \alpha s_j \left[\tfrac{1}{1-\rho} - \tfrac{\rho}{1-\rho} s_{j|g} - s_j \right], \quad \Delta_{jk}^{\text{same}} = \alpha s_j \left[-\tfrac{\rho}{1-\rho} s_{k|g} - s_k \right], \quad \Delta_{jk}^{\text{diff}} = -\alpha s_j s_k$$

Marginal costs are averaged per ASIN across pre-merger quarters and held fixed in both counterfactuals. Holding costs at pre-merger values serves two purposes: it embodies the standard no-efficiencies assumption of prospective merger review, and it avoids a mechanical tautology – if costs were re-inverted in post-merger quarters, the no-merger Nash equilibrium would recover observed prices by construction, since the first-order condition is the identity that defined those costs.

Two Nash equilibria are then computed for every post-merger quarter via a damped fixed-point iteration on the Bertrand first-order conditions:

$$\mathbf{p}_{n+1} = \lambda \left[\mathbf{mc} - (\Omega \circ \Delta(\mathbf{p}_n))^{-1}\mathbf{s}(\mathbf{p}_n)\right] + (1 - \lambda)\, \mathbf{p}_n$$

with $\lambda = 0.4$, convergence tolerance 10^{-8}, and up to 2,000 iterations. The no-merger counterfactual uses pre-merger ownership Ω_{pre}; the merger simulation replaces it with Ω_{post} (hello ASINs assigned to Colgate's firm identifier from 2020Q1). The merger price effect is the difference between the two equilibria, $\Delta p = p^{\text{merger}} - p^{\text{no-merger}}$, so that ownership structure is the only thing that varies between them.

Algorithm validation. The R implementation mirrors the counterfactual equilibrium routines in PyBLP (Conlon & Gortmaker 2020) and is validated against PyBLP's `compute_prices()` directly: for both M1 (plain logit) and M3 (CLIP-nested, $\rho^* = 0.67$), the maximum absolute price difference across 847 post-merger observations is below 2×10^{-8}, confirming algorithmic equivalence.

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5 Results

5.1 Summary Statistics by Cluster

Table 2 reports summary statistics by CLIP cluster. The clusters recover economically meaningful segments without using any sales or price information: a premium sensitivity segment (C1, dominated by the GSK brands), the two mainstream incumbents in their own clusters (C2 Colgate, C3 Crest), Tom's of Maine (C4), and a natural/specialty segment (C5) containing hello. Price levels and package sizes differ sharply across clusters, consistent with the clusters capturing vertical and horizontal differentiation simultaneously.

Table 2: Summary Statistics by CLIP Cluster

			Price (USD)			Mkt. Share (x1000)			Pkg. Size (g)		
Cluster	Dom. Brand	N ASINs	$\bar{p}$	$\tilde{p}$	σ_p	$\bar{s} \times 10^3$	$\tilde{s} \times 10^3$	$\sigma_s \times 10^3$	$\bar{w}$	$\tilde{w}$	σ_w
C1	Sensodyne	29	10.26	9.63	5.58	1.068	0.770	0.848	249	249	145
C2	Colgate	33	8.00	8.39	3.65	1.303	0.817	1.259	537	399	296
C3	Crest	29	8.72	6.88	4.18	1.267	0.757	1.282	375	431	128
C4	Tom's of Maine	13	9.85	9.76	2.96	1.170	0.770	0.972	374	370	94
C5	Other (hello)	55	9.46	7.86	6.84	0.839	0.670	0.736	171	141	105

Notes:
Panel: 1,164 ASIN x quarter observations, 19 quarters (2018Q1–2022Q3). Mean/Median/SD of price in USD, market share scaled x1000, package weight in grams. Dominant brand = modal brand by ASIN count within cluster.

5.2 Demand Estimates

Table 3 presents the demand parameters for all three models, separating the calibrated α from the estimated $\boldsymbol{\beta}$ and the profiled ρ^*. Brand nesting yields $\rho^* = 0.35$ with a 3.1% RSS improvement over plain logit; CLIP nesting yields $\rho^* = 0.67$ with an 8.1% improvement. The stronger profiled nesting parameter indicates that within-cluster substitution is substantially more intense when clusters are defined in embedding space than when they are defined by brand identity.

Table 3: Demand Parameter Estimates — Three Models

Variable	M1: Logit (ρ=0)	M2: Brand-Nested (ρ^*=0.35)	M3: CLIP-Nested (ρ^*=0.67)
α (price)	**−0.310 (a)**	**−0.310 (a)**	**−0.310 (a)**
Package size (g)	0.0037*** (0.0003)	0.0037*** (0.0003)	0.0034*** (0.0003)
CLIP PC_1	-0.4034*** (0.0424)	-0.2869*** (0.0418)	-0.3280*** (0.0407)
CLIP PC_2	0.2033*** (0.0374)	0.2681*** (0.0368)	0.2103*** (0.0359)
CLIP PC_3	-0.1667*** (0.0386)	-0.1479*** (0.0380)	-0.2854*** (0.0370)
CLIP PC_4	0.2115*** (0.0399)	0.1574*** (0.0393)	0.2707*** (0.0382)
CLIP PC_5	-0.1652*** (0.0390)	-0.1243*** (0.0384)	-0.1745*** (0.0374)
Other brand	0.0849 (0.1327)	0.4590*** (0.1307)	0.2641** (0.1272)
Import x freight shock	0.6767*** (0.2342)	0.6709*** (0.2306)	0.7461*** (0.2245)
ρ^*	0.000 (b)	0.350 (b)	0.670 (b)
Implied median ε	−2.62	≈ -4.0	≈ -7.9
RSS drop vs ρ=0	—	3.1%	8.1%

Notes:
*** p<0.01, ** p<0.05, * p<0.10. FEs by quarter. N=1,164. (a) calibrated alpha; (b) grid-searched rho*. Sections 4.2-4.3.

5.3 Own-Price Elasticities by Brand

Table 4 reports median own-price elasticities by brand group for each model. Two patterns are worth noting. First, within each model, elasticities scale with price levels – the nested-logit own-price elasticity is proportional to αp_j, so premium brands such as APAGARD and SENSODYNE PRONAMEL are mechanically the most elastic. Second, and more informative, moving from M1 to M3 amplifies all elasticities by roughly $1/(1-\rho^*)$, but the amplification interacts with cluster composition: brands sharing a crowded embedding cluster face the strongest within-nest competition. The merging parties illustrate the point – hello's median elasticity rises from -1.42 under plain logit to -4.05 under CLIP nesting, reflecting the intense substitution among the natural/specialty products in its cluster rather than substitution toward Colgate.

Table 4: Median Own-Price Elasticities by Brand Group

Brand	N obs.	M1: Logit	M2: Brand-Nested
APAGARD	44	-4.92	-6.62
Sensodyne	99	-3.00	-4.27
SENSODYNE PRONAMEL	90	-3.04	-4.42
Colgate	**224**	**-2.84**	**-4.27**
Tom's of Maine	85	-3.03	-4.38
Other	220	-2.70	-4.09
Crest	219	-2.11	-3.18
Parodontax	51	-1.79	-2.54
Arm & Hammer	21	-1.53	-1.62
hello	**50**	**-1.42**	**-1.88**
JASON	29	-1.30	-1.48
Orajel	32	-0.86	-1.16

Notes:

Median own-price elasticity across all ASIN x quarter observations of each brand group, full estimation panel (1,164 observations). F

5.4 Merger Counterfactuals

Table 5 reports the merger counterfactuals: observed prices, recovered marginal costs, the no-merger Nash equilibrium, the merger Nash equilibrium, and the implied price effect Δp for the merging parties and the largest rivals. Under plain logit, where diversion is proportional to market shares, the simulated effect for hello is +1.23%; brand nesting barely changes it (+1.28%) because hello's ASINs sit in their own brand nest and diversion to Colgate is still share-proportional across nests. Under CLIP nesting, hello (cluster C5) and Colgate (cluster C2) are distant in product space, diversion between the parties is small, and the simulated hello effect falls to +0.40%, with Colgate's own effect economically negligible.

Table 5: Merger Counterfactuals: hello-to-Colgate Acquisition (2020Q1)

Brand	Obs. p	MC	Nash (pre)	Nash (post)	Δp	Δp %
M1: Logit (ρ=0)						
hello	**6.43**	**3.20**	**7.10**	**7.16**	**0.064**	**1.233%**
Colgate	**8.45**	**5.16**	**8.29**	**8.30**	**0.007**	**0.104%**
Tom's of Maine	10.47	7.17	10.70	10.71	0.006	0.069%
Crest	8.87	5.60	9.18	9.18	0.000	0.000%
Sensodyne	10.45	7.17	10.05	10.05	0.000	0.000%
M2: Brand-Nested (ρ^*=0.35)						
hello	**6.43**	**3.20**	**7.10**	**7.17**	**0.065**	**1.275%**
Colgate	**8.45**	**5.16**	**8.30**	**8.30**	**0.006**	**0.104%**
Tom's of Maine	10.47	7.17	10.71	10.71	0.006	0.069%
Crest	8.87	5.60	9.18	9.18	0.000	0.000%
Sensodyne	10.45	7.17	10.07	10.07	0.000	0.000%
M3: CLIP-Nested (ρ^*=0.67)						
hello	**6.43**	**5.20**	**7.05**	**7.07**	**0.020**	**0.404%**
Colgate	**8.45**	**5.88**	**8.59**	**8.59**	**0.002**	**0.018%**
Tom's of Maine	10.47	7.19	10.72	10.72	0.002	0.021%
Crest	8.87	5.63	9.18	9.18	0.000	0.000%
Sensodyne	10.45	7.28	10.29	10.29	0.000	0.000%

Notes:

All prices USD; post-merger averages (2020Q1–2022Q3). Bold = merger parties. Nash (pre): Bertrand-Nash under pre-merger ownership. Nash (post): equilibrium under merged ownership. Dp = Nash (post) minus Nash (pre). MC inverted from pre-merger FOC and held fixed.

5.5 Cross-Model Comparison

The central economic-plausibility comparison across the three models is summarised below. The share of observations with **negative implied marginal costs** is the key diagnostic: when the demand system understates substitutability, the Bertrand inversion attributes implausibly large markups, pushing recovered costs below zero. CLIP-based nesting reduces the negative-MC rate from 8.6% to 1.7% — a five-fold improvement over both the plain logit and brand nesting — while simultaneously delivering the largest RSS improvement.

Table 6: Cross-Model Comparison: Demand Fit and Economic Plausibility

Model	ρ^*	Med. ε_{jj}	RSS drop	Neg. MC %	Med. Lerner	hello Δp	Colgate Δp
M1: Logit	0.00	-2.62	—	8.6%	38.5%	+1.23%	+0.104%
M2: Brand-Nested	0.35	-3.83	3.1%	8.6%	38.5%	+1.27%	+0.104%
M3: CLIP-Nested	**0.67**	**-7.31**	**8.1%**	**1.7%**	**28.9%**	**+0.40%**	**+0.018%**

Notes:
All statistics over the full estimation panel (1,164 ASIN x quarter observations) except merger effects (post-merger quarters 2020Q1–2022Q3). Med. elasticity = median own-price elasticity. RSS drop = reduction in residual sum of squares vs rho=0 from the concentrated OLS profile. Neg. MC = share of observations with negative implied marginal cost from the Bertrand inversion (lower = more economically plausible). Med. Lerner = median (p - mc)/p. hello / Colgate Dp = average post-merger Nash price effect. Bold/green row = preferred specification.

5.6 Figures

Figure 2 visualises the CLIP product space: hello's listings sit in the natural/specialty cluster on the left of the embedding plane while Colgate's cluster occupies the opposite region, making the low implied diversion between the merging parties directly visible. Figure 3 traces the post-merger window quarter by quarter, showing the merger Nash equilibrium diverging from the no-merger counterfactual for hello while the two paths for Colgate remain indistinguishable.

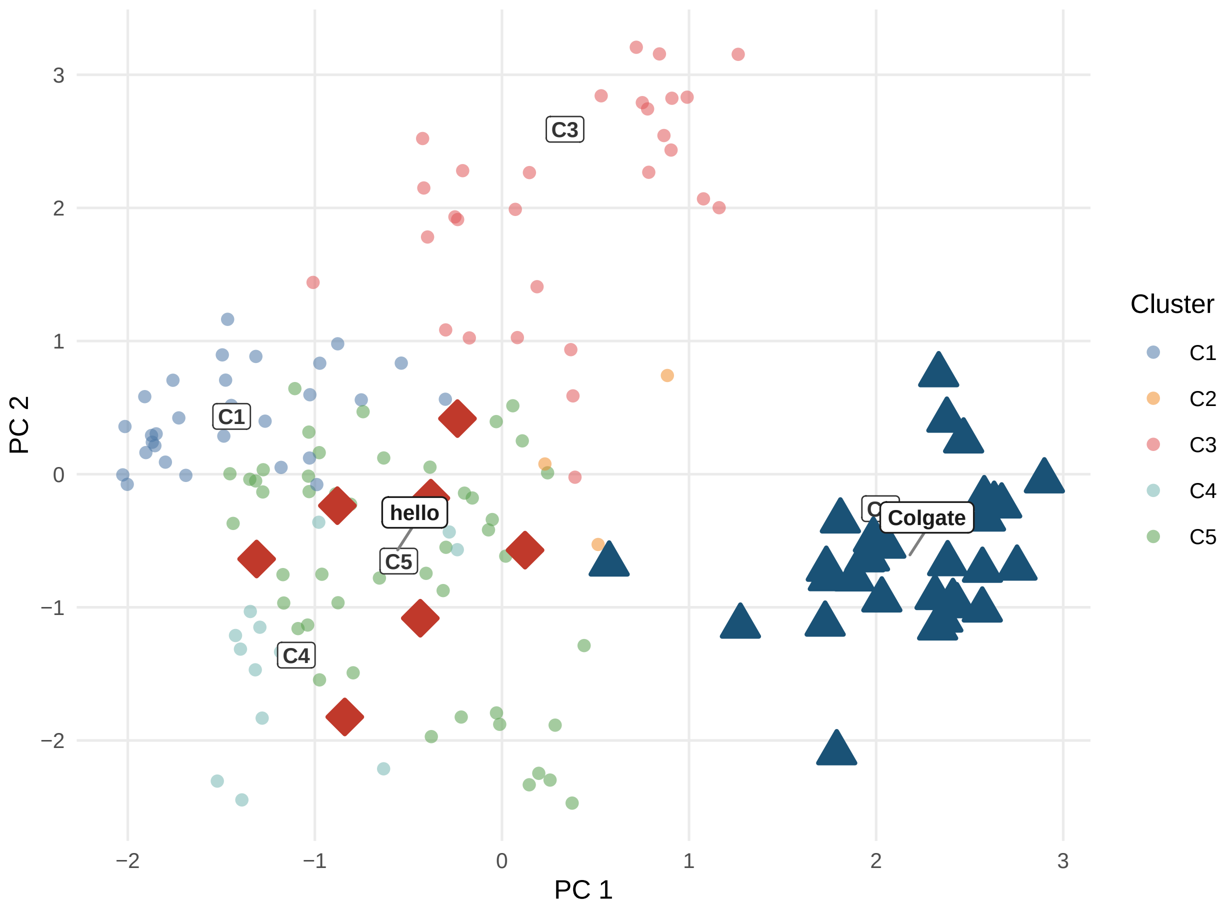

Figure 2: Product space from CLIP embeddings. Each point is one ASIN positioned by its first two principal components of the joint image+text embedding. Cluster centroids are labelled C1–C5. hello (red diamonds, C5) and Colgate (dark blue triangles, C2) occupy different clusters, implying a lower diversion ratio and a smaller predicted merger effect under CLIP nesting relative to plain logit.

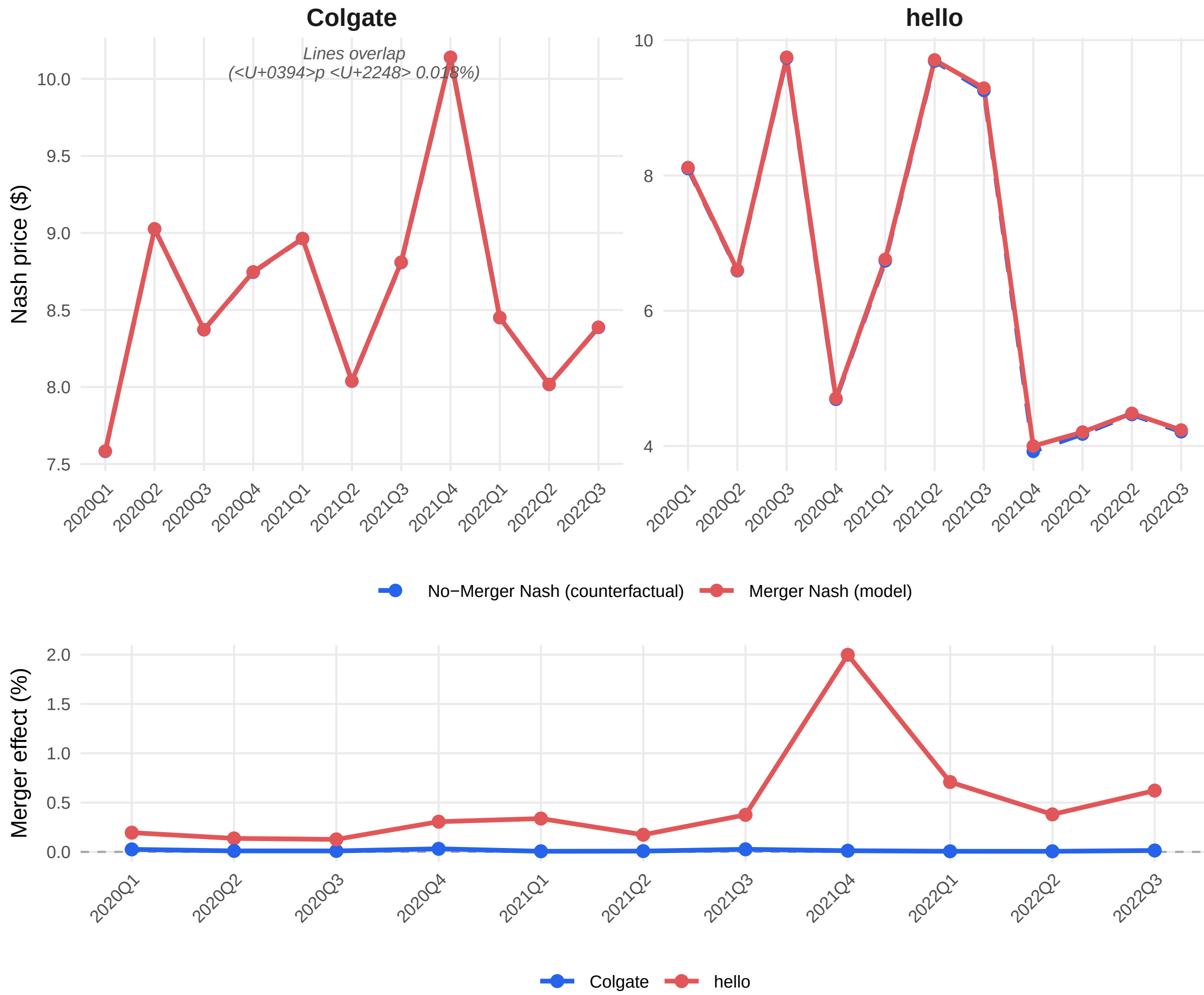

Figure 3: Merger simulation price trajectories (Model 3, CLIP-Nested, $\rho^* = 0.67$). Top panel: Nash equilibrium prices under merged and pre-merger ownership. Observed prices are omitted from the top panel because their quarter-to-quarter variation (\$4–\$10) would compress the counterfactual divergence. Bottom panel: merger price effect in percent, $(p^{\text{merger}} - p^{\text{no-merger}})/p^{\text{no-merger}} \times 100$. hello (red) shows a positive and time-varying effect; Colgate (blue) remains near zero.

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6 Discussion and Limitations

The findings should be read with the design's limitations in view; each is a deliberate trade-off rather than an oversight, and each points to a concrete extension.

Calibrated price coefficient. Valid cost-side instruments are unavailable in this high-frequency online retail panel, so α is calibrated to an external meta-analytic elasticity rather than estimated. The choice makes the price-sensitivity assumption transparent, but it also means α is common across products and the level of all markups inherits the calibration. Because α is calibrated at $\rho = 0$, the nested models imply more elastic demand at their profiled ρ^*; the cross-model *comparison* of fit, negative-cost rates, and merger effects is unaffected, since all three models share the same α. A natural extension is to construct differentiation instruments from the CLIP embeddings themselves – rival products' embedding distances are plausibly excluded from a product's own demand shock – and estimate α rather than calibrate it.

Demand-side cost recovery. Marginal costs are recovered solely from the demand system and the Bertrand first-order conditions; without a supply-side cost equation, true cost variation cannot be separated from demand shocks absorbed into the inversion. The negative-cost rate is used here precisely as a *diagnostic* of demand misspecification, which is robust to this concern, but the recovered cost levels should not be interpreted structurally.

Fixed marginal costs and no efficiencies. The counterfactuals hold costs at pre-merger levels, isolating the market-power effect of the ownership change. If the acquisition generated real marginal-cost efficiencies, the simulated price effects are upper bounds for the merged firm's products.

Static embeddings and cluster count. Embeddings are extracted once per product, which is appropriate for listings that change little, but would miss repositioning. The number of clusters is selected at $K = 5$; Appendix A shows the qualitative conclusions – high ρ^*, low negative-cost rates, small hello effects – are stable across $K = 2$ to 10, though point estimates move.

External validity. The analysis covers one category on one platform around one merger. The pipeline, however, requires only listing content, purchase counts, and prices – data available for essentially any e-commerce category – which is what makes the approach portable to other merger settings.

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7 Conclusion

This paper provides evidence that substitution patterns constructed from unstructured product content outperform those built from brand labels in a workhorse merger-simulation framework. Replacing brand nests with CLIP-embedding clusters in a Berry (1994) nested logit raises demand fit, cuts the rate of economically implausible negative implied marginal costs from 8.6% to 1.7%, and changes the antitrust answer: the simulated price effect of the hello–Colgate acquisition for hello falls from +1.23% under plain logit to +0.40% under CLIP nesting, because the embeddings reveal that the merging parties occupy different regions of product space and divert little demand to one another.

The broader message for practice is that the inputs to credible merger simulation need not be limited to structured characteristics data. Product images and listing text – observable for any online market at essentially zero cost – carry the differentiation consumers act on, and a pre-trained encoder plus standard clustering suffices to put that information into the tools antitrust economists already use. The entire analysis, from embeddings to Nash counterfactuals, is reproducible from a single document and a public repository, which is intended to make the method auditable in exactly the settings where it would be applied.

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9 Appendix

9.1 A: K Sensitivity Analysis

Table 7: K Sensitivity Analysis: CLIP Cluster Count K = 2–10

K	WCSS	Silhouette	ρ^*	RSS Drop	Neg. MC%	hello Δp%	Colgate Δp%
2	1042.0	0.199	0.60	6.1%	0.5%	1.71%	0.015%
3	836.7	0.256	0.66	7.8%	0.9%	3.49%	0.013%
4	641.2	0.330	0.57	5.7%	3.0%	3.96%	0.041%
5	**456.5**	**0.388**	**0.67**	**8.1%**	**1.7%**	**0.40%**	**0.018%**
6	365.6	0.387	0.60	6.7%	3.4%	0.59%	0.044%
7	322.0	0.368	0.48	4.6%	4.4%	0.69%	0.055%
8	279.3	0.400	0.37	2.9%	4.7%	0.82%	0.067%
9	247.7	0.403	0.35	2.4%	4.7%	1.26%	0.177%
10	230.7	0.373	0.40	3.8%	4.6%	0.79%	0.065%

Notes: K-means on 5 CLIP PCs (seed 42, 50 restarts). WCSS = within-cluster sum of squares. Silhouette = average silhouette width (higher = better separation). ρ^* = argmin RSS on grid [0, 0.01, . . . , 0.90]. RSS drop = $100 \times (RSS_0 - RSS^*)/RSS_0$. Neg. MC = share of ASINs with negative implied marginal cost. hello / Colgate Δp = average post-merger price effect in percent (2020Q1–2022Q3). **Bold/green row = selected specification (K=5).**

9.2 B: Model Fit — RMSE and MAE

Table 8: Model Fit: RMSE / MAE vs. Observed Prices (Post-Merger Quarters 2020Q1–2022Q3)

	RMSE / MAE (USD)		
Brand / Scope	M1: Logit	M2: Brand-Nested	M3: CLIP-Nested
All	**1.558 / 0.743**	**1.556 / 0.746**	**1.564 / 0.810**
hello	1.579 / 0.836	1.580 / 0.837	1.533 / 0.799
Colgate	1.144 / 0.559	1.142 / 0.560	1.149 / 0.652

Notes: Each cell shows RMSE / MAE in USD against observed transaction prices. Nash (merger) prices are used as the model prediction. "All" = all 847 ASIN × quarter observations in the post-merger period (2020Q1–2022Q3); hello = 35 obs.; Colgate = 172 obs. Pre-merger marginal costs are held fixed in both Nash simulations.